

02ELMA - Homeworks

① $\vec{\nabla} \times (\vec{\nabla} T) = 0$
 $\vec{\nabla} \cdot (\vec{\nabla} \times \vec{V}) = 0$ } Prove them.

② $\vec{\nabla} T = \frac{\partial T}{\partial r} \hat{r} + \frac{1}{r} \frac{\partial T}{\partial \theta} \hat{\theta} + \frac{1}{r \sin \theta} \frac{\partial T}{\partial \phi} \hat{\phi}$
 $\vec{\nabla} \cdot \vec{V} = \frac{1}{r^2} \frac{\partial}{\partial r} (r^2 v_r) + \frac{1}{r \sin \theta} \frac{\partial}{\partial \theta} (\sin \theta v_\theta) + \frac{1}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi}$ } ~~Spherical coordinates~~
 Spherical coordinates

$\vec{\nabla} T = \frac{\partial T}{\partial s} \hat{s} + \frac{1}{s} \frac{\partial T}{\partial \phi} \hat{\phi} + \frac{\partial T}{\partial z} \hat{z}$
 $\vec{\nabla} \cdot \vec{V} = \frac{1}{s} \frac{\partial}{\partial s} (s v_s) + \frac{1}{s} \frac{\partial v_\phi}{\partial \phi} + \frac{\partial v_z}{\partial z}$ } Cylindrical coordinates

Show that how gradient and ~~divergence~~ is written in spherical and cylindrical coordinates.

③ Calculate $\vec{\nabla} \cdot \vec{E}$ directly from $\vec{E}(\vec{r}) = \frac{1}{4\pi\epsilon_0} \int_{\text{all space}} \frac{\hat{z}}{z^2} \rho(\vec{r}') d\tau'$ and show it is equal to ρ/ϵ_0 .

④ Prove explicitly $\vec{\nabla} \times \vec{E} = 0$ by studying a point charge at the origin
 $\vec{E} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} \hat{r}$ and considering $\oint_{\vec{a}} \vec{E} \cdot d\vec{\ell}$.

⑤ Solve Problem 4.11. (Introduction to Electrodynamics by David J. Griffiths, 3rd Edition) After that assume that ρ_f is the only source for $\vec{D}$. What happens to $\vec{E}$ inside and outside? Is that reasonable?

⑥ Show that $\oint (\hat{r} \cdot \vec{r}') d\vec{\ell}' = -\hat{r} \times \int d\vec{a}'$ (Page 244 in the textbook)

⑦ Prove that $\int \vec{\nabla}' \times \left[\frac{\vec{M}(\vec{r}')}{z} \right] d\tau' = -\oint \frac{1}{z} [\vec{M}(\vec{r}') \times d\vec{a}']$ (See Prob. 1.60, it is related with divergence theorem)